1. A KNN classifier has excellent performance on the training data but noticeably worse performance on new data. Which change would be most likely to help?
    a. Decrease k
    b. Increase k
    c. Give variables with larger variances more weight
    d. Include an interaction term

2. A student says, "The most flexible model is always preferable because it can represent more complicated relationships." Explain what is wrong with this reasoning.

3. You fit Models A, B, and C:

Model	**Training accuracy**	**Testing accuracy**
A	71%	70%
B	86%	82%
C	100%	68%

Which model would you prefer based only on this information? What appears to be happening with Model C?

4. A student makes these three claims:
    a. A predictor with zero Pearson correlation with the outcome cannot improve prediction.
    b. Two highly correlated predictors necessarily interact.
    c. A model with lower training error is always preferable.

    For each claim, state whether it is true or false and explain.

5. A new observation has the following seven nearest neighbors, already ordered from closest to farthest:

Neighbor	1	2	3	4	5	6	7
Class	A	B	B	A	A	B	B

    Predict the new observation's class using $k = 1$, 3, 5, and 7.

6. A student standardizes age but forgets to standardize annual income before fitting KNN. Explain why this is a problem even though both variables are numeric.

7. A researcher wants to use KNN to predict whether or not a bank customer will default on a loan. They write the following code:

```
loan_recipe <-
  recipe(defaulted ~ ., data = loans) |>
  step_dummy(all_nominal_predictors(), one_hot = TRUE)

loan_model <-
  nearest_neighbor(neighbors = 5) |>
  set_engine("kknn") |>
  set_mode("classification")

loan_fit <-
  loan_model |>
  fit(defaulted ~ ., data = loans)
```

Their data contain the following variables:
application_id
age
income
employment_type
credit_score
default_date
defaulted

Identify at least four problems with this analysis. For each problem, explain why it matters and how you would correct it.

8. A scatterplot of X and Y shows a very strong U-shaped relationship. Their Pearson correlation is 0.03. Which statement is best?
    a. X and Y are essentially unrelated
    b. The scatterplot is likely wrong
    c. There is little linear correlation, despite a strong nonlinear relationship
    d. X will not be a useful variable for predicting Y
    e. Both C and D

9. Which EDA finding is evidence of an **interaction**?
    a. X1 and X2 have a Pearson correlation of 0.91
    b. The relationship between X1 and Y is positive for one level of X2 and negative for the other
    c. X1 is strongly right-skewed and X2 is strongly left-skewed
    d. X1 and Y have a Pearson correlation of 0.80
    e. The relationship between X1 and Y is the same across both levels of X2

For Questions 10–12, suppose a university wants to predict whether a student will graduate within four years, using information available **at the end of the student's first year**.

Variable	Description
StudentID	Unique identifier
HS_GPA	High school GPA
FirstYearGPA	GPA at end of first year
Major	Major at end of first year
CreditsAttempted	First-year credits attempted
CreditsCompleted	First-year credits completed
Dorm	First-year residence hall name
GraduationDate	Date of graduation, if graduated
FinalGPA	GPA upon leaving the university
Graduated4Years	Outcome

10. Which variables clearly should **not** be included as predictors in the model? Select all that apply.
    a. StudentID
    b. HS_GPA
    c. FirstYearGPA
    d. GraduationDate
    e. FinalGPA
    f. Major
    g. Graduated4Years
    h. CreditsAttempted
    i. CreditsCompleted

11. Which predictors would require dummy encoding before fitting a logistic regression? Select all that apply.
    a. HS_GPA
    b. FirstYearGPA
    c. Major
    d. CreditsCompleted
    e. Dorm

12. A student creates a new variable, CompletionRate. CompletionRate is calculated as the ratio of CreditsCompleted to CreditsAttempted. Would including CompletionRate as a predictor constitute target leakage? Explain.

13. A student wants to predict “salary” using “experience” and “department.” They fit the following model:

```
salary_recipe <- recipe(salary ~ experience + department,
                        data = employees) %>%
  step_dummy(department, one_hot = TRUE)

salary_model <- linear_reg() %>%
  set_engine("lm")

salary_workflow <- workflow() %>%
  add_recipe(salary_recipe) %>%
  add_model(salary_model)

salary_fit <- fit(salary_workflow, data = employees)
```

Suppose "department" has three levels, "Sales," "IT," and "HR."

After fitting the model, R reports that the model matrix is rank-deficient and that one or more coefficients cannot be uniquely identified.

a. What caused the problem?
b. Why does "one_hot = TRUE" create this problem for an ordinary linear regression model with an intercept?
c. How should the code be changed?
d. After making this change, how many indicator variables will be created for "department"?

14. A logistic regression model predicts whether a patient has pneumonia based on symptoms and clinical measurements. For one patient, the model produces a predicted probability of 0.78. Which interpretation is correct?
    a. The model is 78% accurate for this patient.
    b. The patient definitely has pneumonia.
    c. The fitted model assigns this patient a probability of 0.78 of belonging to the pneumonia class.
    d. 78% of the patient's symptoms indicate pneumonia.

15. A model predicts whether borrowers will default on a loan. Four borrowers receive the following predicted probabilities:

Borrower	**Predicted Probability**
A	.31
B	.48
C	.63
D	.81

a. Using a threshold of 0.50, which borrowers would be classified as likely to default?
b. Using a threshold of 0.70, which borrowers would be classified as likely to default?
c. Explain what happens to the number of positive classifications when the threshold is increased.

16. A logistic regression predicts whether a customer will cancel a subscription. The coefficient for the predictor MonthlyFee is 0.04.

    A student writes: "A $1 increase in the monthly fee causes a customer's probability of canceling to increase by 4%."

    Identify **one problem** with this interpretation and provide a better interpretation.

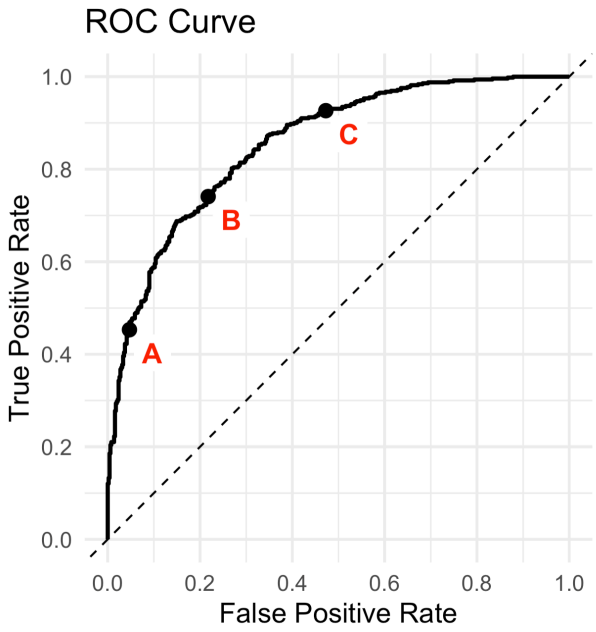
17. A model produces the following confusion matrix:

	Actually Positive	**Actually Negative**
Predicted Positive	72	18
Predicted Negative	28	182

Calculate accuracy, true positive rate, and false positive rate.

18. A student says: "I changed my classification threshold from 0.50 to 0.30, so the ROC AUC of my fitted model will change." Is this correct? Explain.

19. Suppose the ROC curve below represents a model used to screen patients for a very dangerous disease. A false negative is considered substantially worse than a false positive. Of points A, B, and C, which threshold would you be most inclined to choose?



a. Point A
b. Point B
c. Point C
d. All points are equally good
e. Not enough information to decide

20. You have 50,000 observations, two predictors, and clear evidence that the covariance structures differ substantially across levels of your outcome variable. Which would you expect to have the greater potential to capture the true decision boundary?
    a. LDA
    b. QDA
    c. Both would do equally well
    d. Neither would do well because they are both parametric models